UME750: Tribology

L	T	P	Cr
3	0	0	3.0

Course Objective: The objective for this course is to develop an understanding of the tribological behavior of different machine elements and tribo-pairs. This course alsointroduces the concept of contacts of solid surfaces, lubricants, Fluid film lubrication, analysis of friction and wear in bearings and other tribological applications.

Syllabus

Introduction: Introduction to Tribology, Interdisciplinary approach, Tribological considerations in the design of machine elements.

Contact of Solid Surfaces: Geometrical properties of surfaces, surface characterization techniques, contact of engineering surfaces- Herzian and non- Hertzian contacts.

Friction: Adhesion and friction, Causes of friction, laws of sliding and rolling friction, friction instability, theories of friction, friction measurement.

Wear: Wear mechanisms, laws of wear, Measurement of wear, Wear reduction methods.
Fluid Film Lubrication: Introduction, Equations of continuity and motion, Reynold equation, lubrication regimes, Stribeck curve, Hydrostatic and hydrodynamic lubrication, Petroff equation.

Lubricants (Flipped learning component): Functions of lubricants, solid lubricants, mineral and synthetic liquid lubricants, electrorheological, magnetorheological and micropolar lubricants, lubricant additives.

Tribology of Bearings: Bearing types and selection, Sliding bearings- idealized bearings, infinitely long and short journal bearings, Design and analysis of ~~finite~~ hydrostatic and hydrodynamic thrust and journal bearing. Tribological considerations in air, magnetic and hybrid bearings.

Miscellaneous tribological applications: Gear metal forming, brakes of automobile, cutting tools, machine tools, IC engines, tribology in micro/nano devices and biological systems, etc.

Research Assignment: Research assignment will constitute collection of data from industry/internet and interpret the latest research on the new topics in tribology. This also includes technical report writing and seminar presentation.

Course Learning Objectives (CLO)

The students will be able to:

1. understand the theories of friction and wear.
2. calculate the topographical or contact parameters of the solid surfaces.
3. understand and perform the basic calculations for fluid film lubrication problems.
4. calculate the load carrying capacity and performance parameters of hydrostatic sliding bearings.
5. address the prevailing tribological issues in the miscellaneous tribosystems, such as braking systems, gears, IC engines, machining, etc.
6. analyze and present the latest research on the various topics of tribology (through research assignment).

Text Books

1. Bhushan, B., Introduction to tribology, John Wiley and Sons, UK (2013).
2. Arnell R. D., Davies P. B., Halling J. and Whomes T. L. , Tribology: Principles and Design Applications, First edition, Springer-verlag (1991).
3. Hirani, H., Fundamentals of Engineering Tribology with Applications, Cambridge University Press (2016).

Reference Books

1. Majumdar B.C., Introduction to Tribology of Bearings, S. Chand Publishing, New Delhi (2010).
2. Sahoo P., Engineering Tribology, PHI, New Delhi (2005).
3. Bowden, F.P., Tabor, D., Friction: Introduction to Tribology, Heinemann Educational Publishers, London (1974).
4. Khonsari, M.M., Booser, E.R., Applied Tribology: Bearing Design and Lubrication, 2nd edition, John Wiley & Sons, Ltd (2008).
5. Hutchings I.M., Tribology: Friction and wear of engineering materials, Edward Arnold, London, (1992).